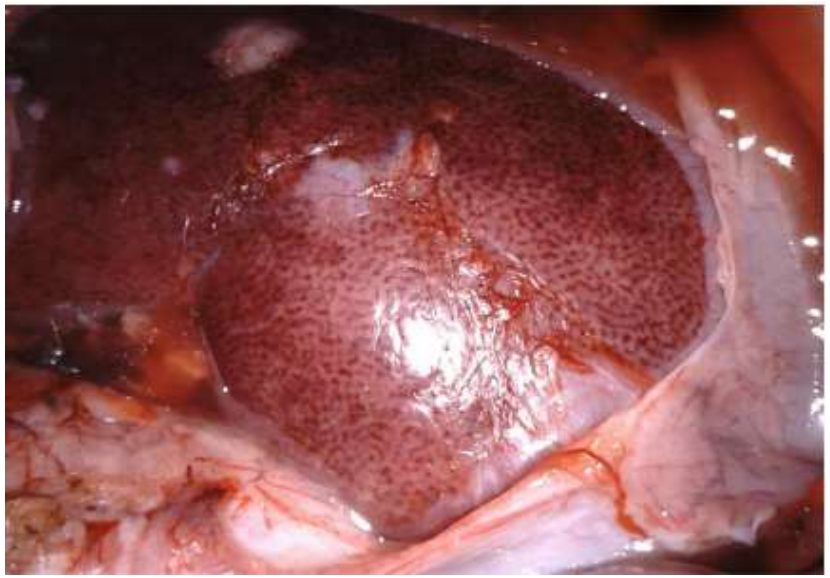
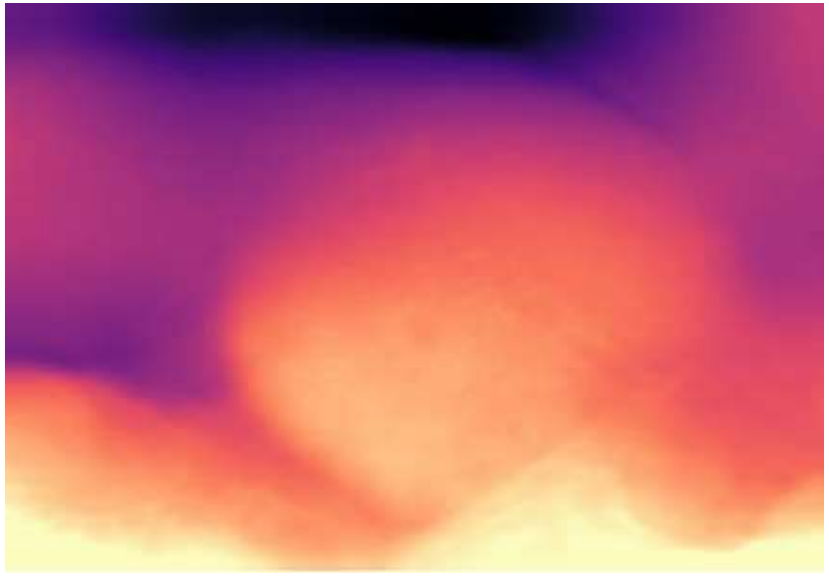
Single-Image Depth Estimation

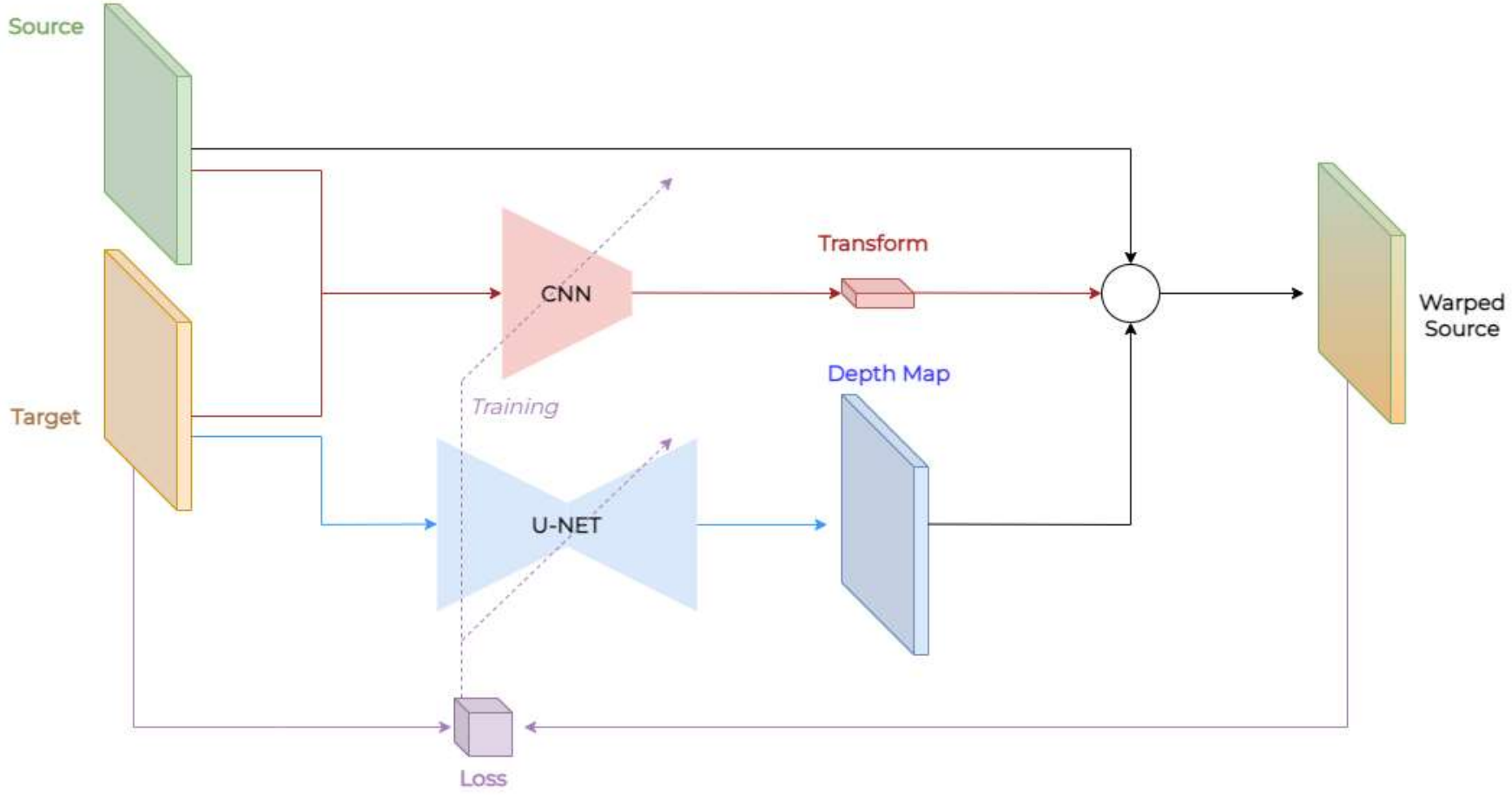
DepthNet & Point Cloud Projections





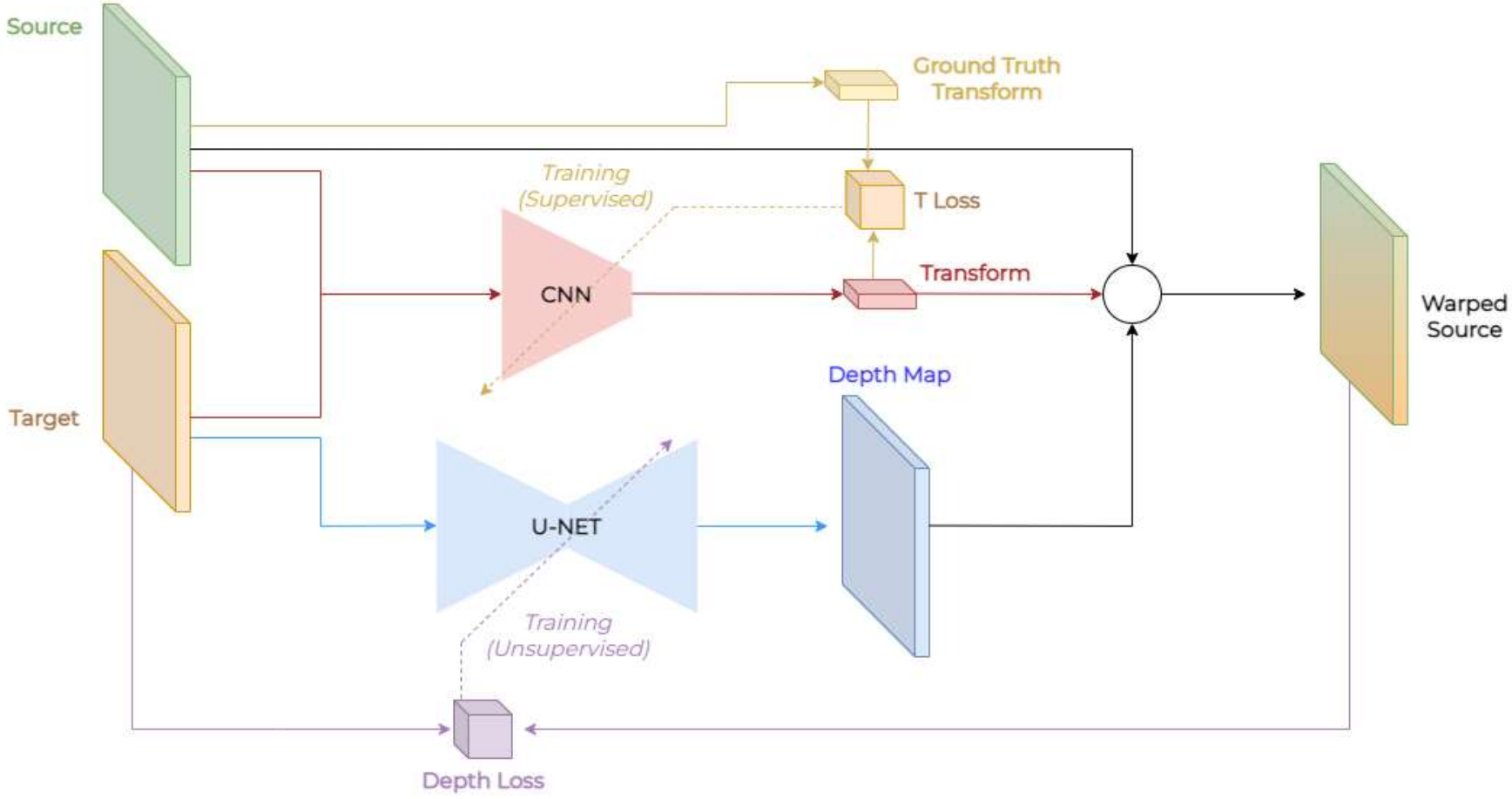
Framework

Training Time



Framework

Training Time

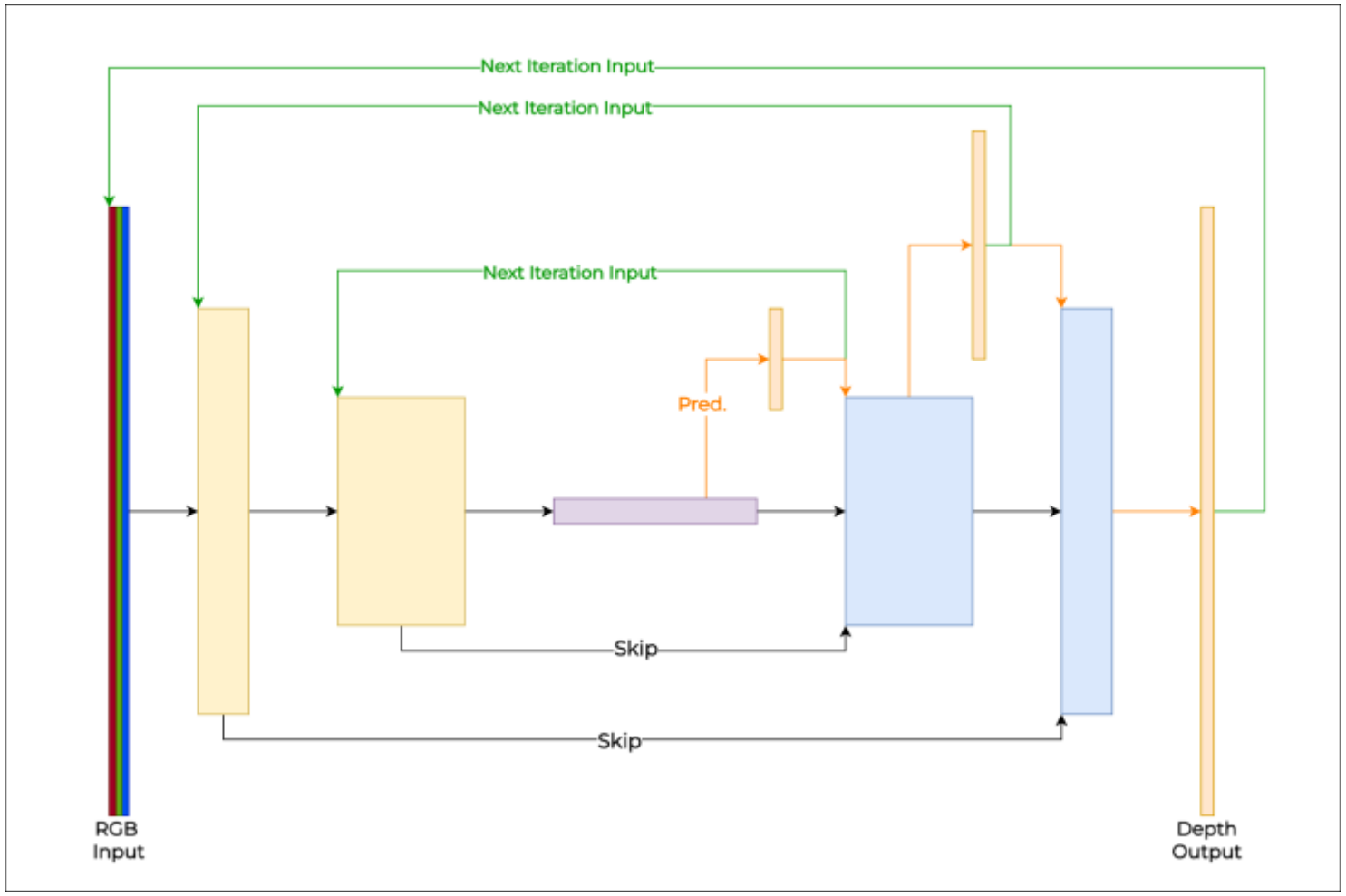


DepthNet

- **UNet**-like architecture
- Multi-scale depth predictions
- Recurrent connections

MODULAR, but

- 6 Conv. Blocks E/D
- **1.7M Params**



Loss Function

- DepthNet is not yet trained → **Random Output** (for now)

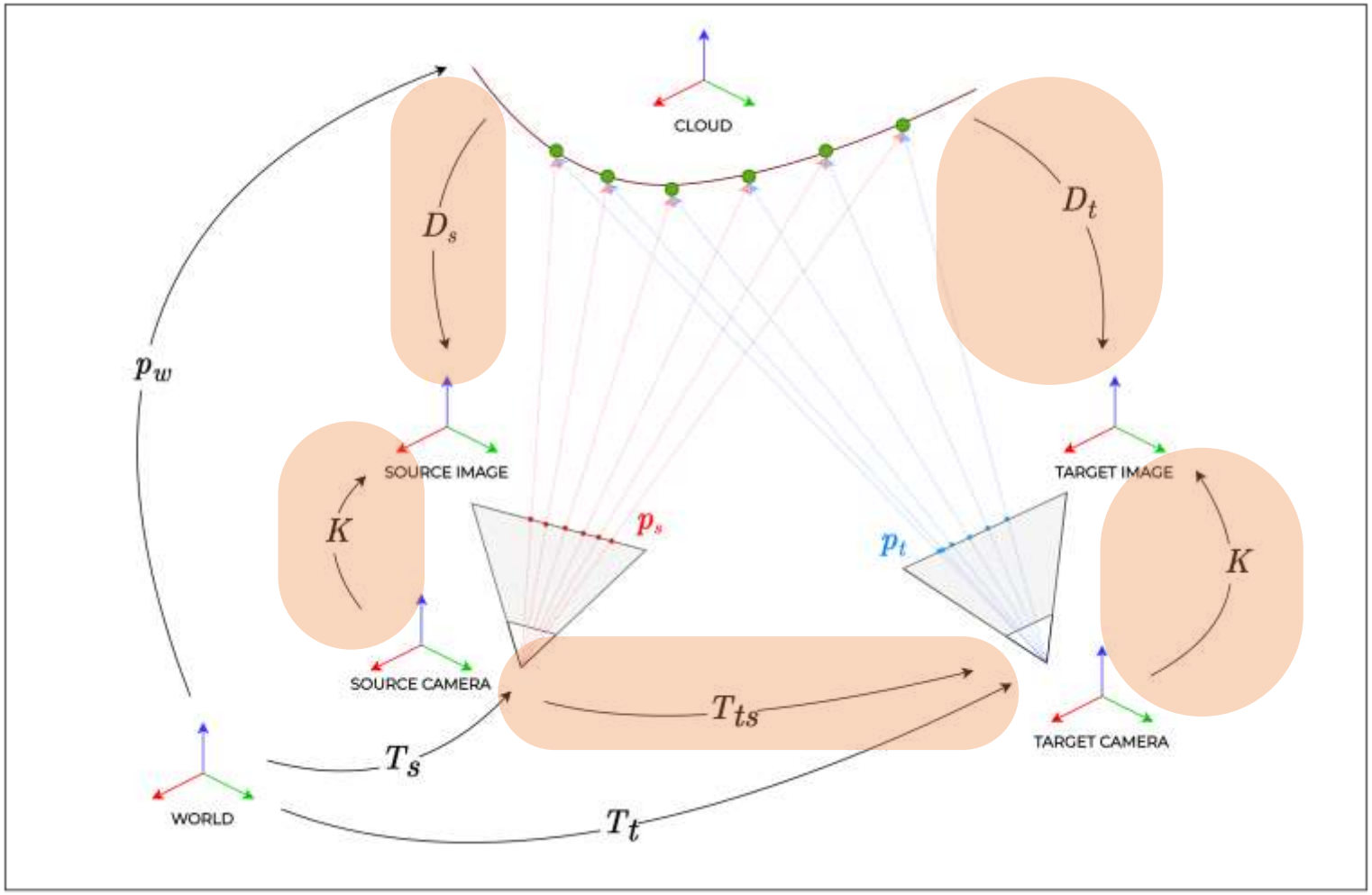
- Supervisory signal → SSIM between **target** frame and **warped-source**

→ Source Warping from
  - Intrinsics *K*
  - Source RGB
  - Source DepthMap
  - Source-To-Target Transform

$$p_t = KT_{s\to t}D_sK^{-1}p_s$$

Errata Corrige from report

Camera Motion Transform Chain



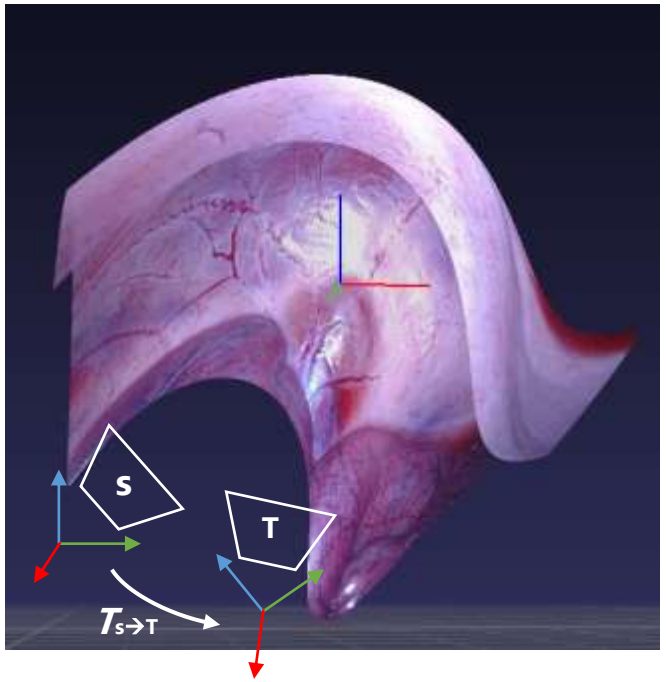
Source-To-Target Warping

Source FRAME

BackProject from Source Frame To Point Cloud

K
DepthMap
RBG

3D Point Cloud



Project from Point Cloud To Target Frame

K
$T_{s\rightarrow T}$

Warped FRAME

Points of Concern

The loss function relies on the computation of the depth map at the previous frame.

To train the network we need either to:

- Train the network with **sequential non-random** sampling from the training dataset
- Use another State Of The Art method to predict the depthmap from the source frame

Which one? Bias? First iteration?

Points of Concern

DepthMap range → How to choose max and min output value?

- A priori / Literature
- From “ground truth data”
- Learned from training

Bias? Can we use stereo algorithm?

→ Sigmoid Mapping from state of the art

Currently Working On

- BackProjection is currently a linear pinhole model

→ Will account for **camera distortion parameters**

- Resolution: Currently X and Y units are $[0; 1]$ normalized

→ Will be converted to **real-world coords** (mm)

- Projection from Point Cloud to Target image